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说明过程各照书里说明估计量=期望ㄑ过猜来.

一. ① 联合分布概率. $Y=(X_1+X_2)^2+(X_3+X_4)^2$

② $\lambda_1=1$ 解=3 ㄑ概率. 指数分布ㄑ无记忆性

二. 极大斯: ① 证 $\alpha=\frac{1}{2}\theta$. 求证 θ 是估ㄑ一半. 正态分布极大斯

三. 离散型: 矩估计. 均方误差. $\begin{cases} P: (X=1) \\ 2P: (X=0) \\ 1-3P: (X=2). \end{cases}$

四. 区间估计. 都布从正态分布.

设假设检验 $X\sim N(\mu,\sigma^2)$ $\ln Y\sim N(\mu,\sigma^2)$.

? 用卡方+卡方做.

五. 秩和检验. $T_1=4$. $T_2=5$. (问题. 两个数不一样)

六. 第一类. 第二类错误. 假设检验 —— 22-23 二题?

七. 单因素方差 —— 题三. 2. P15.

八. ① $Y=\beta_0+\beta_1 X$. 最小二乘估计. β

21-22 三题. ② $\geq$ <2 假设检验

书 P110

一. $Y=(X_1+X_2)^2+(X_3+X_4)^2$. 求 $f(x)$. $\lambda>0$ 时 $P\{Y>32 \mid Y>\lambda\}$

二. $(\theta-2\alpha)^2$ 求证 θ 是估ㄑ一半

三. $P(X=0)=2P$. $P(X=1)=P$. $P(X=2)=1-3P$ ① 矩估计. ② $\hat{p}$ 均方误差

四. $X\sim N(\mu,\sigma^2)$. $\ln Y\sim N(1,\sigma^2)$. μ. σ^2 未知. $X_1\cdots X_7$.

$Y_1\cdots Y_5$. 求 σ^2 区间

六. $X_1\cdots X_{16}$ 来自 $X\sim N(\mu,4)$. $H_0: \mu=3$. $H_1: \mu=4$.

H_0 否定域为 $W=\{(x_1\cdots x_{16}): \bar{x}>3.8\}$.

① 两类错误概率 α 和 β. ② $\alpha=0.01$ 求 $H_0: \mu=3$. $H_1: \mu=4$ ㄑ拒绝域

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$Y=\begin{bmatrix}Y_1\\Y_2\end{bmatrix}=\begin{bmatrix}1&1&0&0\\0&0&1&1\end{bmatrix}\begin{bmatrix}X_1\\X_2\\X_3\\X_4\end{bmatrix}=AX.\sim N(0,\begin{bmatrix}1&1&0&0\\0&0&1&1\end{bmatrix}\begin{bmatrix}X_1\\X_2\\X_3\\X_4\end{bmatrix}\begin{bmatrix}1&0\\1&0\\0&1\\0&1\end{bmatrix})$

$Cov(Y_1,Y_2)=0.$

$Cov(X_1+X_2,\ X_3+X_4)$

$=Cov(X_1,X_3+X_4)+Cov(X_2,X_3+X_4)$

$=Cov(X_1,X_3)+Cov(X_1,X_4)+Cov(X_2,X_3)+Cov(X_2,X_4)$

$=0$

一. (1) X为正态分布$\sim N(0,1)$. $X_1.X_2.X_3.X_4$是X的样本. 求概率密度函数

$Y=(X_1+X_2)^2+(X_3+X_4)^2$

$\therefore (X_1+X_2)$ 与 (X_3+X_4) 独立.

(2) 求 $P\{Y>3\lambda \mid Y>\lambda\}$.

解: (1) $\because X\sim N(0,1)$. $X_1+X_2\sim N(0,2)$. $X_3+X_4\sim N(0,2)$

$\frac{X_1+X_2}{\sqrt{2}}\sim N(0,1)$ $\frac{X_3+X_4}{\sqrt{2}}\sim N(0,1)$ $\frac{1}{2}(X_1+X_2)^2\sim \chi^2(1)$.

$\frac{1}{2}(X_3+X_4)^2\sim \chi^2(1)$ $\checkmark$ $\frac{1}{2}[(X_1+X_2)^2+(X_3+X_4)^2]\sim \chi^2(2).\sim \Gamma(1,\frac{1}{2})$

$Y=(X_1+X_2)^2+(X_3+X_4)^2\sim \Gamma(1,\frac{1}{4})$ 服从指数分布.

$\therefore f(x)=\begin{cases}\frac{1}{4}e^{-\frac{1}{4}x}, & x>0.\\ 0, & x\le 0.\end{cases}$

(2) 根据指数分布的无记忆性 $P\{X>s+t\mid X>s\}=P\{X>t\}$ 得

$P\{Y>3\lambda\mid Y>\lambda\}=P\{Y>2\lambda\}=P\{Y>2\times\frac{1}{4}\}=\int_{\frac{1}{2}}^{+\infty}f(x)dx$

$=\int_{\frac{1}{2}}^{+\infty}\frac{1}{4}e^{-\frac{1}{4}x}dx=-e^{-\frac{1}{4}x}\Big|_{\frac{1}{2}}^{+\infty}=0-(-e^{-\frac{1}{8}})=e^{-\frac{1}{8}}$

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二. 设样本 $X_1 \cdots X_n$ 来自正态总体 $N(\theta, \sigma^2)$，参数 θ 的先验分布为 $N(a_0, \sigma_0^2)$，$L(\theta, d) = (\theta - 2d(x))^2$

① 证明 $\hat{d} = \frac{1}{2}E(\theta|x)$　② 求 θ 的贝叶斯估计.

证明：① 记 $X = (X_1 \cdots X_n)'$，$R(d) = E(L(\theta, d)) = E[\theta - 2d(x)]^2 = E(E\{[\theta - 2d(x)]^2|x\})$

若使 $R(d)$ 最小，则

$E\{[\theta - 2d(x)]^2|x\} = E(\theta^2|x) - 4d(x)E(\theta|x) + 4[d(x)]^2$ 取最小. P69

等式右侧对 d 求导并令其 $=0$，解得 $d(x) = \frac{1}{2}E(\theta|x)$

即当 $d(x) = \frac{1}{2}E(\theta|x)$ 时，$R(d)$ 达到最小.

∴ θ 的贝叶斯估计量为 $\hat{d}(x) = \frac{1}{2}E(\theta|x)$

② $h(\theta|x) = \frac{\pi(\theta) f(x_1 \cdots x_n|\theta)}{g(x)} \propto \pi(\theta) f(x|\theta)$ P76

$$\propto e^{\frac{-(\theta - a_0)^2}{2\sigma_0^2}} \cdot \prod_{i=1}^{n} e^{-\frac{(x_i - \theta)^2}{2\sigma^2}}$$

$$\propto \exp\left\{-\frac{\sigma^2 + n\sigma_0^2}{2\sigma_0^2\sigma^2}\left[\theta - \frac{a_0\sigma^2 + n\bar{x}\sigma_0^2}{\sigma^2 + n\sigma_0^2}\right]^2\right\}$$

∴ $\theta|x \sim N\left(\underbrace{\frac{a_0\sigma^2 + n\bar{x}\sigma_0^2}{\sigma^2 + n\sigma_0^2}}_{E}, \underbrace{\frac{\sigma_0^2\sigma^2}{\sigma^2 + n\sigma_0^2}}_{D}\right)$

∴ $\hat{\theta} = \frac{1}{2}E(\theta|x) = \frac{1}{2} \times \frac{a_0\sigma^2 + n\bar{x}\sigma_0^2}{\sigma^2 + n\sigma_0^2}$

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三.

X	0	1	2
P	2ρ	ρ	$1-3\rho$

① 求 ρ 矩估计　② 求均方误差

解：(1) $E(X)=0\times 2\rho+1\times\rho+2\times(1-3\rho)=2-5\rho$　　$E(X)=\bar{x}$

$\therefore 2-5\hat{\rho}=\bar{x}$ 得 $\hat{\rho}=\frac{1}{5}(2-\bar{x})$

(2) 均方误差 $MSE(\hat{\rho})=E(\hat{\rho}-\rho)^2=D(\hat{\rho})+[E(\hat{\rho})-\rho]^2$

$E(\hat{\rho})=E[\frac{1}{5}(2-\bar{x})]=\frac{2}{5}-\frac{1}{5}E(\bar{x})=\frac{2}{5}-\frac{1}{5}(2-5\rho)=\rho$

$\therefore MSE(\hat{\rho})=D(\hat{\rho})=D[\frac{1}{5}(2-\bar{x})]=\frac{1}{25}D(\bar{x})$

$D(\bar{x})=\frac{1}{n}D(x)=\frac{1}{5}D(x)$.　　$D(x)=E(x^2)-E^2(x)$

其中 $E(x^2)=0\times 2\rho+1^2\times\rho+2^2(1-3\rho)=4-11\rho$

$E(x)^2=(2-5\rho)^2=4+25\rho^2-20\rho$

$\therefore D(x)=9\rho-25\rho^2$　　$D(\bar{x})=\frac{1}{5}(9\rho-25\rho^2)$

$\therefore MSE(\hat{\rho})=\frac{2}{25}\rho-\frac{1}{5}\rho^2$

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四. $X\sim N(\mu,\sigma^2)$. $\ln Y\sim N(1,\sigma^2)$. μ与σ^2均未知. $X_1\cdots X_7$ $Y_1\cdots Y_5$. 求σ^2的区间.

解: ∵ $X_1\cdots X_7$相互独立. $\frac{(n-1)S^2}{\sigma^2}\sim\chi^2(n-1)$ ∴ $\frac{6S^2}{\sigma^2}\sim\chi^2(6)$ ①

$Y_1\cdots Y_5$相互独立. ∴ $\ln Y_1\cdots\ln Y_5$相互独立.

$\sum_{i=1}^{5}\ln Y_i\sim N(5,5\sigma^2)$ $\frac{\sum\ln Y_i-5}{\sqrt{5\sigma^2}}\sim N(0,1)$

∴ $\frac{(\sum\ln Y_i-5)^2}{5\sigma^2}\sim\chi^2(1)$ ②

①+② $\frac{6S^2+\frac{1}{5}(\sum_{i=1}^{5}\ln Y_i-5)^2}{\sigma^2}\sim\chi^2(6+1)$

当$\alpha=0.05$时. $\frac{\alpha}{2}=0.025$. $1-\frac{\alpha}{2}=0.975$.

查表查$\chi^2_{0.975}(7)$ = 1.69 查表查$\chi^2_{0.025}(7)$ = 16.013

∴ $\chi^2_{1-\frac{\alpha}{2}}(7)<\frac{6S^2+\frac{1}{5}(\sum_{i=1}^{5}\ln Y_i-5)^2}{\sigma^2}<\chi^2_{\frac{\alpha}{2}}(7)$

∴ σ^2区间为 $\left(\frac{6S^2+\frac{1}{5}(\sum\ln Y_i-5)^2}{\chi^2_{0.025}(7)},\ \frac{6S^2+\frac{1}{5}(\sum\ln Y_i-5)^2}{\chi^2_{0.975}(7)}\right)$

$\ln Y\sim N(1,\sigma^2)$ $\frac{\ln Y-1}{\sigma}\sim N(0,1)$ $\frac{\sum_{i=1}^{n=5}(\ln Y-1)^2}{\sigma^2}\sim\chi^2(5)$.

①+② $\frac{6S^2+\sum_{i=1}^{n=5}(\ln Y-1)^2}{\sigma^2}\sim\chi^2(6+5)\Rightarrow\chi^2_{1-\frac{\alpha}{2}}(11)<\cdots<\chi^2_{\frac{\alpha}{2}}(11)$

3.816. 21.92.

六. 设 $x_1, \cdots x_{16}$ 来自 $X \sim N(\mu, 4)$ 的简单随机样本. 考虑假设检验问题. $H_0: \mu = 3$. $H_1: \mu = 4$. H_0 的拒绝域为 $W = \{\bar{x} > 3.8\}$

(1) 求犯两类错误的概率 α 和 β.

(2) 在显著性水平 $\alpha = 0.01$ 下求上述拒绝域.

解: (1) $X \sim N(\mu, 4)$. $\bar{x} \sim N(\mu, \frac{1}{4})$ $\frac{\bar{x}-\mu}{1/2} \sim N(0,1)$

当 H_0 成立时. $\mu = 3$. $2(\bar{x}-3) \sim N(0,1)$

$\therefore \alpha = P_0\{\bar{x} > 3.8\} = P_0\{2(\bar{x}-3) > 2(3.8-3)\} = P_0\{2(\bar{x}-3) > 1.6\}$

$= 1 - \Phi(1.6) = 1 - 0.9452 = 0.0548$

当 H_1 成立时. $\mu = 4$. $2(\bar{x}-4) \sim N(0,1)$

$\therefore \beta = P_1\{\bar{x} \leq 3.8\} = P_1\{2(\bar{x}-4) \leq 2(3.8-4)\} = P_1\{2(\bar{x}-4) \leq -0.4\}$

$= \Phi(-0.4) = 1 - \Phi(0.4) = 1 - 0.6554 = 0.3446$

$\therefore \alpha = 0.0548$. $\beta = 0.3446$.

(2) H_0 拒绝域应为 $\{\bar{x} > C\}$

$\alpha = P_0\{\frac{\bar{x}-3}{1/2} > \frac{C-3}{1/2}\} = 1 - \Phi(\frac{C-3}{1/2})$. $C = 3 + \frac{2u_{0.01}}{4}$

$\therefore H_0$ 拒绝域为 $\{\frac{\bar{x}-3}{1/2} > u_{0.01}\} = \{2(\bar{x}-3) > 2.326\}$

八. (2). 对显著性水平 α. 求假设检验 $H_0: \beta_1 \geq 2$. $H_1: \beta_1 < 2$ 拒绝域.

解: $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{L_{xx}})$

H_0 拒绝域为 $\{\hat{\beta}_1 < C\}$ 若 H_0 为真. $\hat{\beta}_1 \sim N(2, \frac{\sigma^2}{L_{xx}})$

则 $\frac{\hat{\beta}_1 - 2}{\sqrt{\frac{\sigma^2}{L_{xx}}}} = \frac{\hat{\beta}_1 - 2}{\sigma}\sqrt{L_{xx}} \sim N(0,1)$

$\therefore \frac{(\hat{\beta}_1 - 2)^2}{\frac{\sigma^2}{L_{xx}}} \sim \chi^2(1)$ $\frac{Q_e}{\sigma^2} \sim \chi^2_{(n-k-1)} = \chi^2(n-2)$

$$F \triangleq \frac{\frac{(\hat{\beta}_1 - 2)^2}{\frac{\sigma^2}{L_{xx}}} / 1}{\frac{Q_e}{\sigma^2} / n-2} \sim F(1, n-2)$$

$\therefore$ 拒绝域为 $\left\{\frac{(\hat{\beta}_1 - 2)^2 L_{xx} \cdot (n-2)}{Q_e} < F_{1-\alpha}(1, n-2)\right\}$

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(2) 若为 $H_0: \beta_0 > 2$, $H_1: \beta_0 < 2$ 求拒绝域.

解: $\hat{\beta}_0 \sim N(\beta_0, \sigma^2(\frac{1}{n}+\frac{\bar{x}^2}{L_{xx}}))$ 继续将 β_0 换为 2. $= (2, \frac{\sigma^2}{n}+\frac{\sigma^2\bar{x}^2}{L_{xx}})$

$$\frac{Q_e}{\sigma^2} \sim \chi^2(n-k-1) = \chi^2(n-2)$$

$$\frac{\hat{\beta}_0-2}{\sqrt{\frac{\sigma^2}{n}+\frac{\sigma^2\bar{x}^2}{L_{xx}}}} \sim N(0,1) \qquad \frac{(\hat{\beta}_0-2)^2}{\frac{\sigma^2}{n}+\frac{\sigma^2\bar{x}^2}{L_{xx}}} \sim \chi^2(1) \qquad F \triangleq \frac{\frac{(\hat{\beta}_0-2)^2}{\frac{\sigma^2}{n}+\frac{\sigma^2\bar{x}^2}{L_{xx}}} \Big/ 1}{\frac{Q_e}{\sigma^2} \Big/ n-2} \sim F(1,n-2)$$

H_0 拒绝域为 $\{F < C\}$.

∴ 拒绝域为 $\left\{ \frac{(\hat{\beta}_0-2)^2 n L_{xx}(n-2)}{(L_{xx}+n\bar{x}^2)Q_e} < F_{1-\alpha}(1,n-2) \right\}$

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